IRIS Flower Classification using Logistic Regression and Random Forest Summer

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1.Abstract

In this project, two machine learning models have been trained and test. Logistic Regression and Random Forest Classifier models have been trained on IRIS Flower Classification data.

2.Introduction

In today's age AI has been rising rapidly in relevance. As such the knowledge and skill to implement machine learning has become valuable. This project helps to familiarize with the basics of machine learning and its implementation.

List of topics on which training was received include Python programming, Logistic Regression, Clustering, Basics of LLMs, Communication skills.

3.Project Objective

This project tries to illustrate the following:

- Splitting a dataset into train and test halves
- Training and evaluating a Logistic Regression model
- Training and evaluating a Random Forest Classifier model

4.Methodology

We train a Logistic Regression model as well as a Random Forest Classifier model. The data for training and testing the models is obtained from the IRIS dataset by importing iris_load from sklearn.datasets. The data is then used to make a DataFrame. From this I made the feature matrix X and independent variable y. Using train_test_split function from sklearn.model_selection X, y are split into testing and training halves with 20% of the data being used for training. Then I made the Logistic Regression model instance and trained it using the data using the fit function. After importing accuracy_score from sklearn.metrics I tested the accuracy of the model's prediction against the test set. Similarly, an instance of a Random Forest Classifier model is made and trained on the data. It is evaluated using classification_report from sklearn.metrics.

5.Data Analysis and Results

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... Logistic regression accuracy:  0.9666666666666667
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The accuracy score of the Logistic regression model is very close to 1.0, this implies the model is very accurate.

	precision	recall	f1-score	support
0	1.00	1.00	1.00	43
1	0.92	0.87	0.89	39
2	0.88	0.92	0.90	38
accuracy			0.93	120
macro avg	0.93	0.93	0.93	120
weighted avg	0.93	0.93	0.93	120

For the Random Forest Classifier the precision and recall of class 0 is the best, implying most accurate predictions for this class. Predictions for classes 1 and 2 are worse than 0 with 2 being marginally better.

6.Conclusion

From this project I have learned how to make and evaluate instances of models as with the Logistic Regression and Random Forest Classifier models.

7.Appendices

1. Github repository link- https://github.com/Orpheus-Mitra/IDEAS-TIH-Internship.git

2. Link for code in Github- https://github.com/Orpheus-Mitra/IDEAS-TIH-Internship/blob/d614ebfd3a2f7008386cb48b88e3ed69d7efc5d5/01_IRIS_Flower_Classification_using_Logistic_Regression_and_Random_Forest_Summer_2026.ipynb